Statistical Analysis Plan

A Randomized, Double-blind, Multi-center, Phase III Study of AK112 or Placebo Combined with Pemetrexed and Carboplatin in Patients with EGFR mutant Locally Advanced or Metastatic Non-squamous NSCLC Who Have Failed to EGFR-TKI Treatment

AK112-301

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Table of Contents

LIST OF **ABBREVIATIONS**.....6
1 OVERVIEW OF THE CLINICAL TRIAL.....8
1.1 OBJECTIVES.....错误！未定义书签。
1.2 ENDPOINTS.....9
1.3 STUDY DESIGN.....10
1.3.1 Study Design.....10
1.3.2 Determination of Sample Size.....11
1.3.3 Randomization and Blinding.....11
2 GENERAL STATISTICAL CONSIDERATIONS.....12
2.1 BASIC PRINCIPLES.....错误！未定义书签。
2.2 ANALYSIS SETS.....12
2.3 MULTIPLE COMPARISONS AND MULTIPLICITY.....12
3 DATA PROCESSING RULES.....13
3.1 DERIVED VARIABLES.....13
3.1.1 Baseline and Change from Baseline.....13
3.1.2 QTcF/QTcB.....13
3.1.3 Study Day.....错误！未定义书签。
3.2 MISSING DATA.....14
3.2.1 Missing Data.....14
3.2.2 Missing/Partial Dates.....14
4 STATISTICAL METHODS.....错误！未定义书签。
4.1 STUDY POPULATION.....16
4.1.1 Dispositions of subjects.....16
4.1.2 Protocol Deviation.....错误！未定义书签。
4.1.3 Demographics and Baseline Characteristics.....16
4.1.4 Prior Cancer Therapy.....错误！未定义书签。
4.1.5 Medical History.....17
4.1.6 Prior and Concomitant Medications.....17
4.1.7 Concomitant Surgery.....18
4.1.8 Concomitant Radiotherapy.....18
4.1.9 Follow-up Cancer therapy.....19
4.1.10 Follow-up Duration.....19
4.2 EFFICACY ANALYSIS.....19
4.2.1 Primary Efficacy Analysis.....19
4.2.2 Secondary Efficacy Analysis.....22
4.3 SAFETY ANALYSIS.....24
4.3.1 Drug Exposure.....错误！未定义书签。
4.3.2 Adverse Events.....25
4.3.3 Clinical Laboratory Evaluation.....30

4.3.4 Vital Signs....32
4.3.5 ECOG.... 32
4.3.6 Electrocardiogram (ECG)....32
4.3.7 Other Safety Analysis.... 33
4.4 IMMUNOGENICITY ANALYSIS....33
4.5 HEALTH-RELATED QUALITY OF LIFE (HRQOL) ANALYSIS....34
4.5.1 EORTC QLQ C30....34
4.6 PHARMACOKINETIC ANALYSIS.... 35
4.6.1 PK Concentration Analysis....35
4.7 BIOMARKER ANALYSIS....36
5 DEVIATIONS FROM PROTOCOL PLANNED ANALYSIS.... 36
6 PLANNED ANALYSIS 36
6.1.1 Periodic Safety Analysis....36
6.1.2 Interim Analysis....36
6.1.3 Final Analysis.... 37
7 STATISTICAL ANALYSIS TABLE/LIST/CHART TEMPLATES....37
APPENDIX INFUSION-RELATED REACTIONS....38

LIST OF ABBREVIATIONS

Abbreviations or specialized terms	Definition
ADA	Anti-drug antibody
AE	Adverse event
AESI	Adverse events of special interest
AUC	scale under the concentration-time curve
CI	Confidence interval
CL	Clearance rate
CR	Complete response
DCR	Disease control rate
DoR	Duration of response
ECG	Electrocardiography
ECOG	Eastern Cooperative Oncology Group
EGFR	Epidermal growth factor receptors
ICF	Informed consent form
irAE	Immun-related adverse events
ITT	Intent-to-treat
MedDRA	International Dictionary of Medical Terms
MSI	Microsatellite instability
NCI CTCAE	National Cancer Institute General Terminology Standard for Adverse Events
NSCLC	Non-small cell lung cancer
ORR	Objective response rate
OS	Overall survival
PD	Disease Progression
PD-1	Programmed cell death-1
PD-L1	Programmed cell death ligand-1
PFS	Progression-free survival
PK	Pharmacokinetics
PKS	Pharmacokinetic analysis set
PR	Partial response
Q3W	Every 3 weeks
QTcB	QT interval corrected by heart rate using Bazett formula
QTcF	QT intervals corrected by heart rate using Fridericia formula

Abbreviations or specialized terms	**Definition**
RECIST	Response evaluation criteria in solid tumors
SAE	Serious adverse event
SUSAR	Suspected and unanticipated serious adverse reactions
SAP	Statistical analysis plan
SD	Stable disease
SID	Subject identification number
SS	Safety Analysis Set
$t_{1/2}$	Half-life
TTR	Time to response
TMB	Tumor mutation burden
TSH	Thyroid stimulating hormone
WHO-DD	WHO Drug Dictionary

1 OVERVIEW OF THE TEST

This Statistical Analysis Plan (SAP) provides a detailed description of the statistical methods and data processing principles for the clinical study "A Randomized, Double-blind, Multi-center, Phase III Study of AK112 or Placebo Combined with Pemetrexed and Carboplatin in Patients with EGFR mutant Locally Advanced or Metastatic Non-squamous NSCLC Who Have Failed to EGFR-TKI Treatment" (Protocol No. AK112-301), which is sponsored by Akeso, Inc..

This SAP is based on the AK112-301 clinical study protocol version 2.0 (version date: 01/18/2022).

1.1 OBJECTIVES

Primary Objectives

- To compare progression-free survival (PFS) assessed by the Independent Radiology Review Committee (IRRC) based on Response Evaluation Criteria in Solid Tumors (RECIST) v1.1, in the mITT population between AK112 combined with pemetrexed and carboplatin and placebo combined with pemetrexed and carboplatin, in patients with locally advanced or metastatic non-squamous NSCLC with EGFR mutations who have progressed on or following EGFR TKI therapy.

Secondary Objectives

- To compare overall survival (OS) in the Modified Intent-to-Treat (mITT) population between AK112 combined with pemetrexed and carboplatin and placebo combined with pemetrexed and carboplatin, in patients with locally advanced or metastatic non-squamous NSCLC with EGFR mutations who have progressed on or following EGFR TKI therapy.
- To compare objective response rate (ORR) and duration of response (DoR), as assessed by IRRC per RECIST v1.1, in the mITT population between AK112 combined with pemetrexed and carboplatin and placebo combined with pemetrexed and carboplatin in patients with locally advanced or metastatic non-squamous NSCLC with EGFR mutation who have progressed on or following EGFR TKI therapy.
- To compare objective response rate (ORR) and duration of response (DoR), as assessed by investigators per RECIST v1.1, in the mITT population between AK112 combined with pemetrexed and carboplatin and placebo combined with pemetrexed and carboplatin in patients with locally advanced or metastatic non-squamous NSCLC with EGFR mutation who have progressed on or following EGFR TKI therapy.
- To compare the safety and tolerability of AK112 combined with pemetrexed and carboplatin and placebo combined with pemetrexed and carboplatin.

- To evaluate the PK profile of AK112 in combination with pemetrexed plus carboplatin.
- To evaluate the immunogenicity of AK112 combined with pemetrexed plus carboplatin.

Exploratory Objectives

- To Explore potential biomarkers in tumor tissue and peripheral blood that may (1) prospectively identify or predict patients most likely to respond to treatment with the study drug and (2) other associations with study treatment response, mechanisms of resistance, and/or disease state;
- To evaluate health-related quality of life (HRQoL) measures of AK112 combined with pemetrexed and carboplatin versus placebo combined with pemetrexed and carboplatin.

1.2 STUDY ENDPOINTS

Primary Endpoints

- PFS assessed by IRRC per RECIST v1.1 based on ITT population.

Secondary endpoint

- OS in ITT population.
- ORR, DoR, DCR and TTR assessed by IRRC per RECIST v1.1 in ITT population.
- PFS, ORR, DoR, DCR, and TTR assessed by the investigators per RECIST v1.1 in ITT population.
- Safety assessment: incidence and severity of adverse events (AEs), clinically significant abnormal laboratory test results.
- Pharmacokinetic Characteristics: AK112 serum drug concentration at different time points after AK112 administration.
- Immunogenicity assessment: number and percentage of patients with detectable anti-AK112 antibody (ADA).

Exploratory Endpoints

- PD-L1 expression in tumor tissue, tumor mutational burden (TMB) and microsatellite instability (MSI) status in tumor tissues.
- Tumor circulating tumor DNA (ctDNA), blood tumor mutational burden (bTMB), and T-cell receptor (TCR) expression levels in peripheral blood.

- HRQoL assessment using the European Organization for Cancer Research and Treatment – Core 30 Questionnaire (EORTC QLQ-C30).

1.3 STUDY DESIGN

1.3.1 Overall Design

This is a randomized, double-blind, multicenter, Phase 3 clinical study. This study plans to enroll about 320 Chinese patients, all of whom have advanced non-squamous NSCLC who have progressed on or following EGFR TKI treatment. This study will evaluate the efficacy and safety of AK112 combined with pemetrexed and carboplatin vs placebo combined with pemetrexed and carboplatin in patients with advanced non-squamous NSCLC who have progressed on or following EGFR TKI therapy.

Based on the patient's previous treatment with a third-generation EGFR TKI (received vs not received) and patient's baseline brain metastases (presence vs absence) as stratification factors, patients will be randomly assigned in a 1:1 ratio to 2 treatment groups, 160 patients each:

- Group A: AK112 (20 mg/kg, Q3W) combined with pemetrexed (500 mg/m2, Q3W) and carboplatin (AUC 5 mg/mL/min, Q3W) for a total of 4 cycles (21 days per cycle), after which AK112 (20 mg/kg, Q3W) and pemetrexed (500 mg/m2, Q3W) will be continued until intolerable toxicity develops or investigator judges there is no longer a clinical benefit (according to RECIST v1.1 using thorough judgment such as
imaging evaluation and clinical condition), or the patient meets other criteria for termination of treatment, whichever occurs first.

- Group B: placebo (20 mg/kg, Q3W) combined with pemetrexed (500 mg/m2, Q3W) and carboplatin (AUC 5 mg/mL/min, Q3W) for a total of 4 cycles (21 days per cycle), after which placebo (Q3W) and pemetrexed (500 mg/m2, Q3W) will be continued until intolerable toxicity develops or investigator judges there is no longer a clinical benefit (according to RECIST v1.1 using thorough judgment such as imaging evaluation and clinical condition), or the patient meets other criteria for termination of treatment, whichever occurs first.

Throughout the study, after a screening period of up to 28 days, all enrolled subjects will receive 4 cycles of AK112 or placebo combined with pemetrexed and carboplatin, followed by maintenance treatment with AK112 or placebo combined with pemetrexed. A PFS interim analysis will be performed after the required number of PFS events for interim analysis has been reached. The interim power of the efficacy data will be reviewed by the Independent Data Monitoring Committee (IDMC). A final PFS analysis will be performed after the required number of PFS events for the final analysis has been reached.

Once enrolled and randomized, patients will have their tumors assessed every 6 weeks (± 7 days) for the first 54 weeks after randomization and every 12 weeks (±7 days) thereafter. OS and PFS assessed by IRRC according to RECIST v1.1 are the primary endpoints. If, after the investigator assesses disease progression based on imaging, the investigator believes the patient will still benefit from continued study treatment (eg, immune-related partial response, immune-related stable disease. and the patient meets the other criteria for continuation in the study, the patient can continue on their assigned study treatment until the investigator deems there is no clinical benefit. If a patient discontinues study treatment for reasons other than disease progression or death, a tumor evaluation should be performed at the EOT visit and tumor evaluations should continue on the same schedule until the patient has progressive disease, starts new antitumor therapy, or exits the study.

The detailed procedures of the study are shown in Tables 7, 8 and 9 of the study protocol.

1.3.2 Determination of sample size

Sample size calculations were based on the number of events required to demonstrate PFS superiority in the AK112 combined pemetrexed and carboplatin group compared over the placebo combined pemetrexed and carboplatin group in the ITT popuklation. The number of events required for PFS was estimated based on the following assumptions (assuming an exponential distribution for event times):

- The one-sided significance level of 0.025;
- The test PFS HR of 0.65 (corresponding to an improvement in median PFS from 6 to 9.2 months) with power of 89%;
- That an interim analysis of the PFS will be performed when information fraction reaches 70%, and the O'Brein-Fleming boundaries were estimated by the Lan-DeMets alpha-spending function;
- That patients will be randomly assigned in a 1:1 ratio to 2 treatment groups; that enrollment will be completed within 12 months with a PFS dropout rate of approximately 5%/12 months

The study plans to enroll approximately 320 subjects, 160 in the experimental group and 160 in the control group. 225 PFS events are expected approximately 21 months after the first subject is randomized, when the final PFS analysis will be performed.

1.3.3 Randomization and blinding

This study will use a stratified block randomization approach, with stratification factors being prior treatment with third-generation EGFR TKI (received vs not received) and baseline brain metastases (presence vs absence). Patients will be randomly assigned to experimental and control groups in a 1:1 ratio.

This study uses a double-blind design. Placebo will be used in the same way as AK112 for both treatment and maintenance periods, and for the double-blind design, the investigator, patient, and other study team members will not know the treatment group of the patient until the study is unblinded.

2 GENERAL STATISTICAL CONSIDERATIONS

2.1 GENERAL STATISTICAL CONSIDERATIONS

All statistical analyses will be done using SAS 9.4 (or later).

They will be summarized and tabulated by treatment group. Descriptive statistics for continuous data will include the number of values, mean, median, standard deviation, minimum, and maximum, while categorical data will be summarized using frequency and percentage.Missing values will not be included in the calculation of percentages unless otherwise stated.

All statistical analyses will be performed at a confidence level of $\alpha = 0.05$ to calculate two-sided 95% confidence intervals (CIs).

2.2 ANALYSIS DATASET

- ITT analysis set: includes all randomized subjects; for analyses requiring baseline data, only subjects with non-missing baseline data will be included (e.g., for ORR, a measurable lesion as defined by RECIST 1.1 is required at baseline). Subjects will be analyzed according to the treatment group they are randomly assigned (regardless of the actual treatment they are receiving).

- Safety Analysis Set (SS): includes all enrolled subjects who received at least one 1 dose of study treatment. The analysis will be based on the treatment patients actually received.

- Pharmacokinetic analysis set (PKS):includes all enrolled patients who received at least 1 dose of AK112 and had at least 1 plasma concentration assessment after dosing .

- The immunogenicity analysis set includes all enrolled patients who received at least 1 dose of AK112 and had predose and at least 1 post-dose immunogenicity assessment.

2.3 MULTIPLE COMPARISONS AND MULTIPLICITY

The primary endpoint in this study only involves comparison between the two treatment groups, so multiple comparisons were not applicable in this trial.

This study has only one primary endpoint of PFS.

For the PFS, in order to strictly limit the global one-sided α at the level of 0.025, the distribution of the α for the PFS interim and final analyses will be calculated using the Lan-DeMets alpha-spending function to estimate the O'Brien-Fleming boundaries.

Analyses of other secondary efficacy endpoints will be treated as supportive and statistical results for these analyses will be interpreted with caution.

3 DATA PROCESSING RULES

3.1 DERIVED VARIABLE

3.1.1 Baseline and change relative to baseline

Baseline is defined as the last non-missing value of the variable taken prior to the first dose of the study drug .

Change from baseline = post-baseline assessment/test value - baseline assessment/test value.

Percent change from baseline = (post-baseline assessment/test value - baseline assessment/test value) / baseline assessment/test value × 100%.

3.1.2 QTcF/QTcB

- QTcF = QT/(RR^0.33) , where RR is a normalized heart rate value and can be obtained by dividing 60 by heart rate.

- QTcB = QT/(RR^0.5) , where RR is the normalized heart rate value and can be obtained by dividing 60 by heart rate.

3.1.3 Study Day

Study day is defined as the number of days to safety / efficacy / pharmacokinetic / immunogenicity assessment from the reference date. The date of the first dose of AK112/placebo will be applied as the reference date. The day of the reference date will be regarded as Day 1.

The study day is calculated as below:

- If the assessment/test is on the same day as or later than the reference date, study day = assessment/test date - reference date + 1;

- If the assessment/test is earlier than the reference date, study day = assessment/test date - reference date.

3.2 MISSING DATA

3.2.1 Missing data

Unless otherwise stated, missing efficacy / safety / pharmacokinetic / immunogenicity data will not be imputed with any hypothetical values.

3.2.2 Missing/incomplete dates

Date of initialdiagnosis

- If only the year of the diagnosis date is recorded, June 30 is applied to impute the missing month and day of the diagnosis date, and if the imputed diagnosis date is later than the date of the first dose, it will be imputed with the date of the first dose.

- If only the year and month of the diagnosis date are recorded, 15th day is applied to impute the missing day in the diagnosis date, and if the imputed diagnosis date is later than the date of the first dose, it will be imputed with the date of the first dose.

- If the date is completely missing, it will not be imputed.

Date of adverse events and concomitant medication

- If only the year and month are recorded for the start date of the adverse event/concomitant medication, and if the year and month are the same as the year and month of date of the first dose of AK112/placebo, the start date of the adverse event/concomitant medication will be imputed with the date of the first dose of AK112/placebo,(if the imputed adverse event/concomitant medication start date is later than the end date, the start date will be imputed directly with the end date) ; otherwise the adverse event/concomitant medication start date will be imputed as the 15th day of the month.

- If only the year is recorded for the adverse event/concomitant medication start date, the adverse event/concomitant medication start date will be imputed with the date of the first dose of AK112/placebo if the year is the same as the year of the date of the first dose of AK112/placebo (if the imputed adverse event/concomitant medication start date is later than the end date, the start date will be imputed with the end date directly); otherwise the adverse event/concomitant medication start date will be imputed as June 30 of the year.

  If the Adverse Event/Combination End Date is incomplete, the Adverse Event/Combination End Date will be imputed as the 15th day of the month in

which the Adverse Event/Combination ended or the last day of the sixth month of the current year (June 30th), but not later than the study end date/date of death (if applicable). If the end date is imputed before the start date, the start date is required.

- If the date is completely missing, it will not be imputed

Date of death

- If only the year of the date of death is recorded and the year is the same as the year of the date of the patient's last known alive day, the patient's last known alive date +1 will be applied to impute the date of death; otherwise June 30 of the year will be applied.

- If only the year and month of the date of death are recorded and the year and month are the same as the year and month of the patient's last known alive date, the date of death was used to populate the date of death by using the patient's last known alive date + 1. Otherwise, the 15th day of the month of the date of death will be applied.

- If the date of death is completely missing, the date of death will be imputed with the patient's last known alive date +1.

New antitumor therapy start date

- If only the year of the new antitumor therapy start date is recorded and the year is the same as the year of the study treatment end date, the study treatment end date + 1 is applied to impute the missing month and day of the new antitumor therapy start date, otherwise the June 30 of the year applied (if the imputed new antitumor therapy start date is later than the end date, the start date will be imputed with the end date directly).

- If only the year and month of the new antitumor therapy start date are recorded and the year and month are the same as the year and month of the study drug treatment end date, the study drug treatment end date + 1 is applied to impute the new antitumor therapy start date, otherwise the 15th of the month is applied (if the imputed new antitumor therapy start date is later than the end date, the start date will be imputed with the end date directly).

- If the new antitumor therapy start date was completely missing, the study treatment end date +1 will be applied to impute the new antitumor therapy start date.

4 STATISTICAL METHODS

4.1 STUDY POPULATION

4.1.1 Disposition of subjects

Subject disposition will be summarized across all screening subjects.

The number of cases and reasons for screening failures and successful screening but not randomized will be summarized separately. And a listing will be provided for screening failures and successful screening but not randomized.

Numbers and percentages of subjects who are randomized, receive at least one study dose, reach the end of treatment, and reach the end of the study will be summarized by treatment groups; And the numbers and percentages of reasons for the end of treatment and the end of the study will be summarized respectively.A listing of the disposition of subjects will also be provided.

The number and percentage of subjects in each analysis data set and reasons for exclusion from the analysis data set are summarized. A listing of subject distribution of each analysis data set will be provided and the reasons for exclusion from certain analysis data sets will be demonstrated.

4.1.2 Protocol deviations

Protocol deviations will be categorized as significant and non-significant.

A listing will be provided to describe significant protocol deviations for subjects in the ITT analysis set; And all protocol deviations will be summarized.

4.1.3 Demographics and baseline characteristics

Based on the ITT analysis set, the demographic and baseline characteristics will be summarized and the data will be listed accordingly.

4.1.3.1 Demographics

Demographics (age and age subgroups (<65 years vs. ≥65 years), sex, ethnicity, height, weight, body mass index (BMI)) will be summarized and tabulated.

4.1.3.2 Baseline disease characteristics

The following baseline disease characteristics will be summarized and tabulated:

- Clinical stage at entry diagnosis (IIIB/IIIC, IV)
- Baseline brain metastases: yes, no
- Baseline liver metastases: yes, no
- Number of distant metastatic lesions (<3, ≥3)

- Smoking history (never smoked, current/former smoker)
- Baseline ECOG score (0, 1)
- Baseline EGFR mutations (19Del, L858R, other)
- T790M mutation status (negative, positive, unknown)
- Prior EGFR-TKI treatment
  - Only have failed first- or second-generation EGFR-TKI
  - have failed 1st or 2nd and 3rd generation EGFR-TKIs
  - have failed only 3-generation EGFR-TKIs

Summary of stratification factors (from IWRS) will be displayed

- Prior treatment with 3rd generation EGFR-TKI (received, not received)
- Baseline brain metastases (present, absent).

4.1.4 Prior cancer therapy

Based on ITT analysis set, the number of cases and percentages of subjects of prior cancer therapy (surgery, radiation therapy, systemic antitumor therapy) will be calculated by treatment groups.

In addition, prior cancer therapies will be coded using WHODrug Global 2022 Mar 1, or later versions, providing the international nonproprietary name (i.e., generic name), preferred names (PN), and anatomical therapeutic chemical (ATC) classification of the drug. Based on the ITT analysis set, the number and percentage of subjects will be calculated by prior cancer therapies categories (TKI class, immune checkpoint inhibitors, chemotherapeutic agent, herbal medicine, etc.) and PN, .

Based on ITT analysis set, the prior cancer therapies will be displayed by the treatment groups.

4.1.5 Medical history

Medical history will be coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 25.0 or higher.

Based on the ITT analysis set, the number and percentage of subjects were calculated by summarizing medical histories by system organ classification (SOC) and preferred terminology (PT).

In addition, the subject's medical history will be tabulated.

4.1.6 Prior and concomitant medications

Prior and concomitant medications will be coded using WHODrug Global 2022 Mar 1, or later versions, to provide the international nonproprietary name (i.e., generic name),

preferred name (PN), and anatomical therapeutic chemical (ATC) classification of the medication.

Based on the ITT analysis set, the number and percentage of subjects were calculated by summarizing prior and concomitant medications according to ATC classification level 2 and PN, respectively:

- Prior medications, i.e., medications that were initiated prior to AK112/placebo and had ended their use prior to the first administration of AK112/placebo;

- Concomitant medications, i.e., medications that start prior to, on, or after the first dose of AK112/placebo, and no later than 90 days after the last dose of AK112/placebo or the start of a new antitumor therapy (whichever occurs first), and end on or after the day of the first dose of AK112/placebo, or that are still ongoing at the end of the study.

If the time of start of a medication relative to the time of the first dose of AK112/placebo could not be determined, the medication will be considered to be a concomitant medication.

A list of prior and concomitant medications for the subject will be provided.

4.1.7 Concomitant surgical treatment

Concomitant surgical treatments will be coded using MedDRA version 25.0 or later. Based on the ITT analysis set, the numbers and percentages of subjects will be calculated by System Organ Classification (SOC) and Preferred Terminology (PT).

Concomitant surgical treatments are defined as treatments started prior to, on, or after the first dose of AK112/placebo, and treatment started no later than 90 days after the last dose of AK112/placebo or the start of a new antitumor therapy (whichever occurs first), and end on or after the day of the first dose of AK112/placebo, or that are still ongoing at the end of the study.

In addition, a list of concomitant non-drug treatments will be provided.

4.1.8 Concomitant radiation therapy

Based on the ITT analysis set, the number and percentages of subjects who received radiation therapy will be summarized. They will also be summarized by indications and location of radiation therapy.

Concomitant radiation therapies are defined as therapies started prior to, on, or after the first dose of AK112/placebo and started no later than 90 days after the last dose of AK112/placebo or the start of a new antitumor therapy (whichever occurs first), and end

on or after the day of the first dose of AK112/placebo or that are still ongoing at the end of the study.

A list of concomitant radiation treatments will also be provided.

4.1.9 Follow-up cancer therapy

Follow-up systemic antitumor therapies will be coded using WHODrug Global 2022 Mar 1, or later version, providing the international nonproprietary name (i.e., generic name), and preferred name (PN) of the medication. Based on the ITT analysis set, the number and percentages of subjects will be summarized by follow-up cancer therapies' drug class (TKI class, immune checkpoint inhibitor, chemotherapeutic agent, herbal medicine, etc.) and PN. In addition, a list of follow-up antitumor treatments (systemic antitumor therapy, surgical therapy, radiation therapy) will be provided.

4.1.10 Follow-up time

Follow-up time is defined as a subject's duration from randomization to the last known alive date or the data cutoff date; if the subject is alive at the time of the data cutoff date, the date of the last known alive will be the cutoff date. If the date of last known subject survival is after the DCO, the date of the last known alive will be the data cutoff date. Specific censoring rules are listed below:

Scenario	Status indicator	Date of event or date of censoring
Alive at the time of data cut-off	Event	last known alive date or data cut-off date
death	Censor	Date of death

The Kaplan-Meier method was used to obtain the survival function estimates and median follow-up time, and the Brookmeyer-Crowley method with Log-Log transformation was used to construct two-sided 95% CIs for the median follow-up time in the ITT population.

4.2 EFFICACY ANALYSIS

Efficacy will be analyzed in the ITT population. Radiology data will be evaluated by IRRC and the investigator, respectively. The primary study endpoint is PFS assessed by IRRC per RECIST v1.1.

4.2.1 Primary efficacy analysis

4.2.1.1 Progression-free survival (PFS) assessed by IRRC per RECIST v1.1

4.2.1.1.1 principal analysis

PFS is defined as the time from randomization until disease progression (as assessed by RECIST v1.1) or death from any cause, whichever occurs first. At the time of analysis,

patients who did not progress or die will be censored on their last evaluable tumor assessment date (or on the date of randomization if patients did not have an evaluable tumor assessment after randomization). The detailed event/censoring scheme for PFS is shown in the table below.

Table 2 Event/Censoring Scheme for PFS/DOR

Scenario	**Status indicator**	**Date of event or date of censoring[2]**
No baseline or post-baseline tumor assessments (No death or death occurs after two missing tumor assessments) [1]	Censor	Date of randomization
Not experiencing any events and not starting new anticancer therapy	Censor	Date of last evaluable tumor assessment
Documented PD	Event	Date of 1st documented PD per RECIST v1.1 criteri
Death without PD	Event	Date of death
Death or PD occurs after two or more consecutive missing tumor assessments[3]	Censor	Date of last evaluable tumor assessment before missing assessment
Starting new anticancer therapy	Censor	Date of last evaluable tumor assessment before starting new anticancer therapy

1. Not applicable to DoR.
2. If ≥2 scan dates were available for an evaluable assessment, the earliest scan date will be used if overall response is PD, and the latest scan date will be used if overall response is PR/CR/SD.
3. For visits during the ≤ x-th week, missing tumor assessments for two or more consecutive times are defined as an interval of more than x1 days between two evaluable tumor assessments. For visits after the x-th week, missing tumor assessments for two or more consecutive times are defined as an interval of more than x2 days between two evaluable tumor assessments. If a subject dies more than x days after randomization and no evaluable tumor assessment is available after baseline, the subject will then be censored on the date of randomization

Missing more than one assessment consecutively is considered if PD or the last evaluable assessment before death (overall efficacy assessment non-NE) is < (48-1) × 7 from the date of the first study drug administration, and PD or death and the date of the last evaluable assessment are separated by > (6+1+6+1) × 7; if PD or the last assessment before death (overall efficacy assessment non-NE) is ≥ (48-1) × 7 but < (54-1) × 7 from the date of first study drug administration, and two or more assessments are considered to be missed if the dates of PD or death and the last evaluable assessment are separated by > (6+1+12+1) × 7; and if the last evaluable assessment before PD or death (overall efficacy assessment non-NE) is separated by > (6+1+12+1) × 7 from the first study drug administration date ≥ (54-1) × 7, and two or more assessments were considered missed if the date of PD or death and the date of the last evaluable assessment were separated by > (12+1+12+1) × 7.

The raw data will keep only subject information prior to and on the cutoff date, so there is no need to make judgments about the order of the cutoff date and the event or censoring date.

Hypothesis testing for the PFS is as follows:

$H_0 : PFS(t)_{experimental} = PFS(t)_{control}$

H_A : $PFS(t)_{experimental} \neq PFS(t)_{control}$

Here PFS(t) denotes the survival function of PFS at time t, $t \geq 0$.

There will be 1 interim analysis and 1 final analysis of PFS in the trial, with one-sidedvα = 0.007 for the 1st interim analysis and α = 0.023 one-sided for the final analysis.

In the ITT analysis set, frequencies and percentages of subjects with PFS events, PD, deaths, and censoring will be summarized for each treatment group. The median, Q1, Q3, minimum, and maximum of PFS time will be estimated using the Kaplan-Meier method and 95% CIs for median, Q1, and Q3 of PFS time will be estimated using the Brookmeyer-Crowley method by Log-log transformation. The progression-free survival rate and its 95% CI at different time points (e.g. at 3, 6, 9, 12 months) will also be reported.

Survival curves between treatment groups were compared using stratified log-rank tests of which the stratification factors are the same as the stratified Cox regression model, and the p-value of the log-rank test will be reported, based on stratification factors reported at randomization. Stratification factors were prior treatment with third-generation EGFR TKI (received vs not received) and baseline brain metastases (presence vs absence). Stratification factors were obtained from the IWRS system if not otherwise stated.

Hazard ratio (experimental group vs control group) and its 95% CI will be estimated from the stratified Cox regression model with the treatment group as an explanatory variable and prior treatment with third-generation EGFR TKI (received vs not received) and baseline brain metastases (presence vs absence) at randomization as stratification factors. The exact method is used to handle the tied events in Cox regression.

The type-I error rate for the PFS interim and final analyses will be adjusted by the number of PFS evaluated by the IRRC events actually obtained from the interim analysis using the Lan-DeMets alpha spending function to estimate O'Brien-Fleming boundaries. Overall one-sided α will be strictly controlled at the 0.025 level.

All PFS information including event/censoring description and date will be provided at subject-level for each treatment group in listings

4.2.1.1.2 sensitivity analysis

If applicable, the following sensitivity analyses will be planned for the PFS endpoint evaluated by the IRRC:

- Sensitivity analysis 1: Analysis was performed using unstratified Log-rank Test and unstratified Cox proportional hazard model.

- Sensitivity analysis 2: If there are subjects whose actual stratification factors recorded in CRFs do not agree with those from IWRS, treat prior treatment with third-generation EGFR TKI (received vs not received) and baseline brain metastases (presence vs absence) at randomization from CRFs as stratification factors, and perform stratified log-rank test statistic for comparing PFS between two treatment groups, as well as obtain the hazard ratio and its 95% CI from the stratified Cox regression model.
- Sensitivity Analysis 3: treat death or disease progression after two or more consecutive tumor assessments as PFS events.
- Sensitivity analysis 4: Disregarding whether the subject used new antitumor therapy prior to PD assessed by IRRC or death, treat death or disease progression after starting new anticancer therapy as PFS events.

4.2.2 Secondary efficacy analysis

4.2.2.1 Overall survival (OS)

Subjects who do not provide any follow-up information will be censored at the randomization date. OS is defined as the time from the date of randomization until the date of death from any cause. Subjects who are alive at the time of analysis will be censored on the last known survival date. Subjects whose last known survival date is beyond DCO will be censored on the DCO instead.

Based on the ITT, OS will be analyzed using statistical methods similar to the primary analysis of PFS endpoint evaluated by the IRRC, frequencies and percentages of subjects with PFS events, PD, deaths, and censoring will be summarized for each treatment group. The median of OS time will be estimated using the Kaplan-Meier method and 95% CIs for median of OS time will be estimated using the Brookmeyer-Crowley method. The progression-free survival rate and its 95% CI at different time points (e.g. at 6, 9, 12, 18 months, etc.) will also be reported. The same stratified log-rank method as the main analysis of PFS assessed by IRRC will also be used to compare between-group differences and to estimate HR and its 95% CI based on the stratified Cox proportional risk model.

4.2.2.2 Objective response rate (ORR) assessed by IRRC per RECIST v1.1 criteria

IRRC-assessed ORR is defined as the proportion of subjects where the IRRC-assessed best overall response (BOR) is complete response (CR) or partial response (PR).

BOR is defined as the best response documented from the first dose of study drug until disease progression (in the order of CR, PR, SD, PD, or not evaluable (NE)), or the best response documented from the first dose of study drug until the last evaluable tumor assessment before the start of subsequent anticancer therapy or the study end, if PD is not achieved (whichever occurs first). Subjects with no post-baseline assessment for any reason will be considered as non-responders for BOR.

ORR assessed by IRRC per RECIST v1.1 will be analyzed based on all ITT subjects who have measurable lesions per RECIST v1.1 criteria at baseline. Frequencies and percentage of subjects in terms of ORR assessed by IRRC per RECIST v1.1 criteria will be calculated, and the 95% CI will be estimated using the Clopper-Pearson method. In addition, different BOR outcomes (including CR, PR, SD, PD, and NE) will be summarized with frequencies and percentages. As for the difference in ORR between two groups, the 95% CIs are obtained from the Miettinen-Nurminen method whereas p-values are obtained from the Cochran-Mantel-Haenszel test, with randomized stratification factors assigned in the IWRS system. In addition, the number of cases and proportion of subjects with various BOR (CR, PR, SD, PD and NE) in each treatment group will be summarized.

4.2.2.3 Duration of Response (DoR) assessed by IRRC per RECIST v1.1 criteria

DoR is defined as the time from the first objective response (CR or PR) to documented disease progression assessed by IRRC per RECIST v1.1 criteria or death from any cause, whichever occurs first Subjects who do not achieve PD or death will be censored at the date of the last tumor assessment. The detailed event/censoring logic for DoR is the same as for PFS, as detailed in Table 2. DoR will only be evaluated for subjects who achieve a confirmed CR or PR.

DoRs assessed by IRRC per RECIST v1.1 will be analyzed using statistical methods similar to those used in the IRRC PFS primary analysis.

4.2.2.4 IRRC Disease Control Rate (DCR) per RECIST v1.1 assessment

DCR is defined as the proportion of subjects who achieve CR, PR, or SD per RECIST v1.1 criteria.

The DCRs assessed by IRRC per RECIST v1.1 will be analyzed using statistical methods similar to those used in the ORR assessed by IRRC per RECIST v1.1.

4.2.2.5 IRRC to mitigation time (TTR) per RECIST v1.1 evaluation

TTR is defined as the time from first dose of study drug until the first objective response (CR or PR). TTR will only be evaluated for subjects with a BOR of CR or PR

Based on ITT, TTRs assessed by IRRC per RECIST v1.1 will be descriptively summarized by treatment group for subjects with a BOR of CR or PR.

4.2.2.6 PFS, ORR, DoR, DCR, TTR assessed by investigators per RECIST v1.1

Statistical analysis methods are the same as IRRC assessed PFS, ORR, DoR, DCR and TTR per RECIST v1.1.

4.2.2.7 Consistency of IRRC and investigator assessments

Based on ITT, Frequencies and percentage of subjects with consistent and inconsistent IRRC and investigator-assessed responses (BOR of CR or PR) and PD will be calculated respectively. In addition, for subjects with both IRRC and investigator assessed documented PD, frequencies and percentage of subjects with a difference of >7 weeks (7-14 weeks, 14-21 weeks, and >21 weeks) between IRRC and investigator assessment dates of PD will also be analyzed.

4.2.2.8 Subgroup analysis

Based on ITT, analyses of the efficacy endpoints PFS and OS will be estimated by treatment group in each category of the following subgroups, respectively:

- Age (<65, ≥65)
- Sex (male, female)
- Clinical stage at entry (IIIB/IIIC, IV)
- Number of distant metastatic sites at baseline (<3, ≥3)
- Baseline liver metastases (present, absent)
- Smoking history (yes (former or current), never)
- Baseline ECOG score (0, 1)
- Baseline EGFR mutations (19Del, L858R, other)
- T790M mutation (negative, positive)
- Number of lines of prior EGFR-TKI treatment
  - First-line (including those who have used only a 1st or 2nd generation EGFR-TKI and those who have used only a 3rd generation EGFR-TKI)
  - Second-line (including with 1st or 2nd and 3rd generation EGFR-TKI)
- Stratification factors (based on actual stratification factors)
  - Baseline brain metastases (present, absent)
  - Prior treatment with 3rd generation EGFR-TKI (received vs. not received)

In addition, if the data are applicable, other subgroups of demographic and baseline characteristics may be analyzed exploratively and will be displayed in forest plots.

For PFS and OS,The unstratified Cox proportional hazards model will be used in subgroup analysis. If population of any subgroup is less than 10% of the ITT, that subgroup will not be shown in the forest plot.

The efficacy endpoints PFS and OS will also be analyzed based on the immunogenicity analysis set, grouped by ADA test results (treatment-emergent ADA-positive, non-treatment-emergent ADA-positive, and ADA-negative).

4.3 SAFETY ANALYSIS

All safety analysis will be based on SS.

4.3.1 Extent of Exposure

AK112/placebo, pemetrexed and carboplatin will be analyzed separately. Total number of doses received by subjects, actual cumulative dose, duration of exposure, actual dose intensity, and relative dose intensity will be summarized for each treatment group using descriptive statistics (mean, median, standard deviation, minimum and maximum). In addition, Subjects with certain durations of exposure (i.e., ≥3 months, ≥6 months, ≥12 months) will also be summarized with frequencies and percentages:

- Actual cumulative dose (mg) = sum of all actual administered doses of study drug;
- Duration of exposure (cycle) = ((date of last dose + 21 days) – date of first dose) / 21, duration of exposure would be truncated to the date of death, date of DCO, or date of last dose + 21 days, whichever occurs first;
- Actual dose intensity (mg/cycle) = actual cumulative dose (mg) / (duration of exposure (cycle)).
- Planned dose intensity (mg/cycle) = planned cumulative dose (mg) / (duration of exposure (days)/ treatment cycle), planned cumulative dose is defined as all the initially planned study drug doses for the duration of exposure.
- Relative dose intensity (%) = actual dose intensity (mg/cycle) / planned dose intensity (mg/cycle) × 100.

Details of extent of exposure will be provided at subject-level for each treatment group in listings.

4.3.2 Adverse event

All AEs will be coded using the MedDRA version 25.0, or later versions.

A treatment-emergent adverse event (TEAE) is defined as an AE that occurred during treatment (i.e., occurred on or after the first dose of study drug to 30 days [for non-SAE] or 90 days [for SAE] after the last dose of study drug, or started new anti-cancer therapy, whichever occurs first), or an existing AE that worsened in severity after treatment in comparison with before treatment. All irAEs occurring up to 90 days after the last dose of study drug will be considered, regardless of starting new anti-cancer therapy.

A treatment-related adverse event (TRAE) is defined as a TEAE with relationship to study drug (including AK112/placebo, pemetrexed, carboplatin), classified as “related”,

"possibly related", "probably related" and "Unlikely related". TEAE with missing relationship to study drug will be considered as TRAE for the sake of conservation .

All TEAEs will be summarized whereas all AEs during the study period will be provided in listings.

4.3.2.1 TEAE Summary Sheet

Frequencies and percentages of subjects will be presented for the following categories of TEAEs:

- All TEAE
  - TEAE related to AK112/placebo
  - TEAE related to pemetrexed
  - TEAE related to carboplatin
- TEAE with CTCAE Grade ≥ 3
  - TEAE related to AK112/placebo with CTCAE grade ≥ 3
  - TEAE related to pemetrexed with CTCAE grade ≥ 3
  - TEAE related to carboplatin with CTCAE grade ≥ 3
- TEAE leading to treatment cycle delay of any study drug
  - TRAE leading AK112/placebo to treatment cycle delay
  - TRAE leading pemetrexed to treatment cycle delay
  - TRAE leading carboplatin to treatment cycle delay
- TEAE leading to treatment discontinuation of any study drug
  - TRAE leading AK112/placebo to treatment discontinuation
  - TRAE leading pemetrexed to treatment discontinuation
  - TRAE leading carboplatin to the treatment discontinuation
- Serious TEAE
  - Serious TEAE related to AK112/placebo
  - Serious TEAE related to pemetrexed
  - Serious TEAE related to carboplatin
- TEAE leading to death
  - TEAE that is not related to disease progression leading to death
  - TEAE related to AK112/placebo leading to death
  - TEAE related to pemetrexed leading to death
  - TEAE related to carboplatin leading to death
- Infusion-related reactions
  - Infusion-related reactions leading AK112/placebo to treatment cycle delay
  - Infusion-related reactions leading AK112/placebo to treatment discontinuation
  - Serious infusion-related reactions
  - Infusion-related reactions leading to death
- Immune-related AE (irAE)
  - CTCAE level 3 or above irAE

- irAE leading to treatment cycle delay of AK112/placebo administration
- irAE leading to treatment discontinuation of AK112/placebo
- Serious irAE
- irAE leading to death

4.3.2.2 TEAE analyzed according to MedDRA SOC and PT

Frequencies and percentages of subjects will be presented by MedDRA SOC and PT for the following categories of TEAE:

- All TEAE (by group PT and PT, respectively)
- TEAE related to AK112/placebo
- TEAE related to pemetrexed
- TEAE related to carboplatin

4.3.2.3 Death, SAE and other important TEAEs

4.3.2.3.1 Deaths

The date of death and cause of death (PD, adverse event, others) will be summarized descriptively:

- All deaths

- On-treatment Deaths:
  - Deaths occurred on or after the first dose of study drug to 30 days after treatment ends, or started new anti-cancer therapy, whichever occurs first;

  - Deaths occurred between 30 days after treatment ends and 90 days after treatment ends, or started new anti-cancer therapy, whichever occurs first;

- Deaths during follow-up, i.e., deaths occurred beyond 90 days after treatment ends, or after starting new anti-cancer therapy, whichever occurs first.

In addition, Frequencies and percentages of subjects will be presented by MedDRA SOC and PT for the following categories of TEAE. Details of death will be provided at subject-level for each treatment group in listings.

- TEAE leading to death
- TEAE related to AK112/placebo leading to death
- TEAE related to pemetrexed leading to death
- TEAE related to carboplatin leading to death

4.3.2.3.2 SAE

Subjects with SAE will be summarized by MedDRA SOC and PT with frequencies and percentages:

- SAE
- SAEs related to AK112/placebo
- SAEs related to pemetrexed
- SAEs related to carboplatin

All SAEs will be provided at subject-level for each treatment group in listings.

4.3.2.3.3 Other significant TEAEs

Subjects with the following significant TEAEs will be summarized by MedDRA SOC and PT with frequencies and percentages:

- TEAE with CTCAE grade ≥3
- TEAE related to AK112/placebo with CTCAE grade ≥3
- TEAE related to pemetrexed with CTCAE grade ≥3
- TEAE related to carboplatin with CTCAE grade ≥3
- TEAE with PT frequencies ≥ 5%
- TEAE related to AK112/placebo with PT frequencies ≥ 5%
- TEAE leading any study drug to treatment discontinuation
- TEAE leading AK112/placebo to treatment discontinuation
- TEAE leading pemetrexed to treatment discontinuation
- TEAE leading carboplatin to treatment discontinuation
- TEAE leading to treatment cycle delay of any medication
- TEAE leading AK112/placebo to treatment cycle delay
- TEAE leading pemetrexed to treatment cycle delay
- TEAE leading carboplatin to treatment cycle delay

4.3.2.4 irAE

Subjects with the following irAE will be summarized by the irAE group defined by sponsor-physician and MedDRA PT, with frequencies and percentages:

- All irAE
- irAE with CTCAE Grade ≥ 3
- irAE leading AK112/placebovto treatment cycle delay
- irAE leading AK112/placebo to treatment discontinuation
- Serious irAE
- irAE leading to death.

Subjects with the irAE listed above will be summarized by the irAE group defined by sponsor-physician, with frequencies and percentages. Descriptive statistics will be provided unless otherwise stated.

- Time to first occurrence of irAE
  - Calculated among subjects with irAE only
  - Defined as time from the first dose of study drug to the first irAE occurrence within the irAE group
- Duration of irAE
  - Calculated among subjects with irAE only
  - When the subject has multiple irAEs, the irAEs will be merged if the lag time in between is ≤3 days within the same irAE group for the same subject. Duration of irAE will then be calculated after merging.
  - Duration of irAE is calculated by the time from the first occurrence of irAE to the final resolution of irAE within the same group. If the irAE is not resolved, the duration will be truncated to the earliest date of [treatment end date + 90 days], DCO, or date of death
- Summary of dose and duration of Systemic Corticosteroid
  - Calculated among subjects with irAE and on systemic corticosteroids only
  - The initial dose of systemic steroids therapy is defined as the sum of the prednisone dose or equivalent dose of all systemic steroids used on the first day for subjects with irAE and on systemic steroids therapy for the first time. Refer to irAE analysis plan for detailed systemic steroid dose conversion.
  - High starting dose is defined as the dose of the first occurrence of prednisone use ≥40mg/day or equivalent dose while on systemic steroids therapy for irAE
  - Treatment duration is defined as the sum of the time of steroids use during the first irAE treated with high-dose steroids, excluding the time of withdrawal and the overlaps when multiple steroids are taken together. If the lag time in between is ≤3 days within the same irAE for the same subject, the treatment duration will be merged. If the steroids treatment is continuing, the duration will be truncated to the earliest date of [treatment end date + 90 days], DCO, or date of death.
- Summary of outcome

- Time to irAE remission
  - Calculated among subjects with irAE and irAE outcome of recovering/resolving (from highest CTCAE grade) only
  - If the lag time between consecutive irAEs is ≤3 days within the same irAE group for the same patient, the irAEs will be counted as one continuous event
  - Time to irAE remission is defined as the longest duration from the initial occurrence to the resolution of the most severe irAE (i.e., irAE with the highest CTCAE grade) within the irAE group. If the same subject experiences multiple irAEs within the same irAE group, the irAE with the highest CTCAE grade will be considered. For irAEs with the same CTCAE grade, the longest time to resolution will be used when all irAEs have resolved. If the most severe irAE has not been resolved, the date of the earliest occurrence of [treatment end date + 90 days], DCO, or date of death will be considered as the censoring date.
- Median, minimum and maximum values will be provided
- Summary of Hormone Replacement Therapies (for Endocrine irAE Subjects only).

All irAE data will be provided at subject-level for each treatment group in listings.

4.3.2.5 Infusion-related reactions

Subjects with infusion-related reaction will be summarized by MedDRA SOC and PT with frequencies and percentages.

All infusion-related reaction data during the study period will be provided at subject-level for each treatment group in listings.

4.3.2.6 Adverse Events of Special Interest (AESI)

Subjects with AESI, AESIs with CTCAE ≥ grade 3 defined by sponsor-physician, will be summarized by MedDRA SOC and PT with frequencies and percentages, respectively.

All AESI data during the study period will be provided at subject-level for each treatment group in listings.

4.3.3 Laboratory tests

Laboratory evaluations include hematology, chemistry, urinalysis, thyroid function, and coagulation profile.

The abnormal laboratory results and toxicity grade will be graded according to NCI CTCAE Version 5.0 as applicable. Shift tables will be presented to compare the changes of CTCAE grades from baseline to the worst post-treatment results. In addition, according to the toxicity terms in CTCAE Version 5.0, frequencies and percentages of subjects with at least one CTCAE Grade worsens, CTCAE Grade worsens 2 levels or above, CTCAE Grade worsens to Grade 3 or Grade 4 post-treatment will be summarized.

Frequencies and percentages of subjects who meet the following pre-defined abnormal criteria of liver function at any visit post-treatment (including unscheduled visits) will be summarized:

- Alanine transaminase (ALT)
  - ≥3×ULN
  - ≥5×ULN
  - ≥10×ULN
  - ≥20×ULN

- Aspartate transaminase (AST)
  - ≥3×ULN
  - ≥5×ULN
  - ≥10×ULN
  - ≥20×ULN

- Transaminases (ALT or AST)
  - ≥3×ULN
  - ≥5×ULN
  - ≥10×ULN
  - ≥20×ULN

- Total bilirubin
  - ≥2×ULN

- Alkaline phosphatase (ALP)
  - ≥1.5×ULN

- Transaminases (ALT or AST) and total bilirubin
  - ALT/AST ≥3×ULN and BILI≥1.5×ULN
  - ALT/AST ≥3×ULN and BILI ≥2×ULN

- Transaminases (ALT or AST) and total bilirubin and alkaline phosphatase (ALP) (Hy's Law)
  - ALT/AST ≥3 × ULN and BILI ≥2 × ULN and ALP <2 × ULN.

Frequencies and percentages of subjects who meet the following pre-defined abnormal criteria of thyroid function at any visit post-treatment (including unscheduled visits) will be presented:

- TSH>ULN
- TSH > ULN and baseline TSH ≤ ULN
- TSH>3×ULN
- TSH > 3 × ULN and baseline TSH ≤ ULN
- TSH>10×ULN
- TSH > 10 × ULN and baseline TSH ≤ ULN
- TSH<LLN
- TSH < LLN and baseline TSH ≥ LLN

All laboratory test results will be listed and values which fall outside the laboratory normal range will be flagged on the data listings. In addition, All laboratory test with CTCAE Grade worsens post-treatment will be provided in listings.

4.3.4 Vital Signs

As for vital signs parameters, including systolic blood pressure, diastolic blood pressure, respiration rate, heart rate, pulse rate, temperature and weight, the observed values and changes from baseline will be summarized using descriptive statistics at each visit, including end-of-treatment visit, visit with maximum value among all visits, visit with minimum value among all visits.

All vital signs data will be provided at visit-level for each treatment group and for each subject in listings.

4.3.5 ECOG

Shift tables will be presented to compare the changes of ECOG score from baseline to the worst post-treatment results by treatment group and visit.

All ECOG data will be provided at visit-level for each treatment group and for each subject in listings.

4.3.6 Electrocardiogram (ECG)

A shift table will be presented to compare the baseline to the worst post-baseline visit, using a categorization of “Normal”, “Abnormal, not clinically significant”, “Abnormal, clinically significant”.

In addition, for QTcF, QTcB and QT, subjects with any one of the following “new abnormalities” at any post-baseline visit (including unscheduled visits) in comparison with baseline measurements will be reported with frequencies and percentages:

- >450 mses
- >480 mses
- >500 mses
- >30 msec increase from baseline
- >60 msec increase from baseline
- >90 msec increase from baseline.

"New abnormality" is defined as a QTc abnormality that was not present at baseline, but emerged at least once in post-baseline ECG assessments.

All ECG results will be provided at visit-level for each treatment group and for each subject in listings.

4.3.7 Other Safety Analysis

All other safety evaluations (e.g. pregnancy test) will be provided in listings.

4.4 IMMUNOGENICITY ANALYSIS

Based on the immunogenicity analysis set, the frequencies and percentages of subjects who are ADA-positive will be summarized by treatment group during the trial, including baseline and post-baseline. In addition, for subjects who are ADA-positive after baseline, the proportion of subjects with ADA produced by treatment and ADA produced by non-treatment will be calculated, and further analysis of neutralizing antibodies (NAb) will be performed for subjects with ADA produced by treatment (if available):

Non-treatment-emergent ADA-positive: Any post-treatment positive ADA assay response that is less than 4-fold over baseline titer level when baseline is ADA positive in the ADA assay.

Treatment-emergent ADA-positive: 1). Any post-treatment positive ADA assay response when the baseline ADA result is negative; 2). Any post-treatment positive ADA assay response that is greater than or equal to 4-fold over baseline titer level when baseline is ADA positive in the ADA assay; 3). Baseline titer level or post-baseline titer level is missing when baseline is ADA positive.

Frequencies and percentages of subjects will be summarized will be summarized based on the immunogenicity analysis set.

In addition, the results of efficacy and safety statistical analysis for different ADA categories will be summarized if the data are applicable.

4.5 HEALTH-RELATED QUALITY OF LIFE (HRQOL) ANALYSIS

HRQoL will be analyzed based on the ITT analysis set.

4.5.1 EORTC QLQ C30

EORTC QLQ-C30 has a total of 30 items, which can be categorized into 15 scales, including:

- 5 functional scales (Physical, Role, Cognitive, Emotional and Social Functioning)
- 3 symptom scales (Fatigue, Pain, Nausea/Vomiting)
- 1 Global Health Status/QoL scale
- 6 single item sacles (each as an individual scale).

The specific scales and scoring rules are listed below:

	Scale	Number of items	Item range	Item numbers
General health status/QoL				
General health status/QoL	QL2	2	6	29, 30
functional scale				
Physical functioning	PF2	5	3	1 to 5
Role functioning	RF2	2	3	6, 7
Emotional functioning	EF	4	3	21~24
Cognitive functioning	CF	2	3	20, 25
Social functioning	SF	2	3	26, 27
Symptom scales / items				
Fatigue	FA	3	3	10, 12, 18
Nausea and vomiting	NV	2	3	14, 15
Pain	PA	2	3	9, 19
Dyspnoea	DY	1	3	8
Insomnia	SL	1	3	11
Appetite loss	AP	1	3	13
Constipation	CO	1	3	16
Diarrhoea	DI	1	3	17

Financial difficulties	FI	1	3	28

For all scales, the RawScore, RS, is the mean of the component items: $RS = (I_1 + I_2 + \cdots + I_n)/n$.

If missing data are encountered when calculating of a raw score, the average of values for the non-missing questions may be used to impute the missing data as long as at least half of the items from the scale have been answered. If the number of missing questions is greater than or equal to half of the total number of questions needed for that raw score, the raw score is set to missing. Scores for single item scales are not going to be imputed if missing.

In order to make the scales comparable with each other, a linear transformation will be used to standardise the raw score, which has a range from 0 to 100:

Standardized score (SS) for the functional scales: $SS=\{1-(RS-1)/range\} \times 100$.

Standardized scores for symptom scales / single items and Glbal health status / QoL $SS = \{(RS-1)/range\} \times 100$.

Scales scores will be summarized by treatment group and visit. Observations and their change from baseline (frequencies of subjects, mean, standard deviation, median, minimum, and maximum) will be reported.

All EORTC QLQ-C30 results will be provided at subject-level for each treatment group and for each visit in listings.

4.6 PHARMACOKINETIC ANALYSIS

4.6.1 Concentration analysis

AK112 serum concentrations will be analyzed based on PKS.

Individual serum concentrations will be listed by scheduled sampling time points. For statistical summary, concentrations below the quantification limit (BQL) will be set to 0 and indicated as “BQL” in the list. Missing data will be considered missing.

Serum concentrations of each individual will be summarized into a table by scheduled sampling time points, including the number of evaluable cases, arithmetic mean, standard deviation, coefficient of variation, minimum value, median value, maximum value, geometric mean, geometric standard deviation, and geometric coefficient of variation. The mean serum concentration-time curves (arithmetic mean + standard deviation) will be plotted using linear and semi-logarithmic scales against the scheduled sampling time points. Sampling time points that exceed the time window will not be included in the quantitative description of drug concentrations or in the individual or mean serum concentration-time curves.

Additionally, if applicable, trough concentrations after multiple doses will be plotted in box plots or linear plots .

4.7 BIOMARKER ANALYSIS

Biomarker analysis will be described in a separate analysis plan document.

5 DEVIATIONS FROM PROTOCOL PLANNED ANALYSIS

This statistical analysis plan makes no changes to the analysis planned in the protocol.

6 CONDUCTING THE ANALYSIS PLAN

6.1.1 Regular safety analysis

Safety data will be analyzed regularly (approximately every 6 months) in this study.

IDMC review meetings will be held regularly (approximately every 6 months) during which the study will continue to enroll subjects. At the regular data review meetings, safety will be assessed by evaluating the data. Following the data review, the IDMC may recommend modifications to the study, including “The clinical trial can be continued without any protocol modifications”, or “The clinical trial can be continued after revising the protocol”, or “Enrollment should be suspended until uncertainties are resolved”, or “The clinical trial should be terminated (due to the observed serious safety concerns)”.

6.1.2 Interim analysis

An interim analysis of PFS is planned for this study; the interim analysis of PFS will be performed when the number of PFS events reaches 70%, approximately 14 months after the first subject is randomized. The O'Brein-Fleming boundaries will be calculated by the Lan-DeMets spending function.

Interim analyses will be conducted by an external IDMC independent statistician. IDMC will review the results for interim analysis. Following the data review, and considering the overall risks and benefits to trial participants, the IDMC will provide recommendations on whether the final analysis can proceed as planned, or whether modifications are needed, or whether the result from interim analysis can be submitted to the regulatory agency.

The stopping boundaries (p-values and Z-scores) for PFS superiority at the time of the interim and final analyses are shown in the table. The boundaries will be updated using an alpha-spending function based on the actual events observed at the interim analysis and the number of events expected for the final analysis

Table 3 Boundaries (p-values and Z-scores) and Approximate Hazard Ratio (HR) for Interim and Final PFS analyses

	Time (months)	#Event	One-sided p-value for efficacy (Z-score)	Approximate HR boundary for efficacy
PFS (ITT analysis set) (one-sided α = 0.025)				
Interim analysis	14	158	0.007 (-2.438)	0.678
Final analysis	21	225	0.023 (-2.0)	0.766

6.1.3 Final analysis

Final PFS analysis will be conducted when 225 PFS events observed in the study.

7 STATISTICAL ANALYSIS TABLE/LIST/CHART TEMPLATES

Templates for statistical analysis tables/lists/charts will be provided separately in a separate document.

APPENDIX INFUSION-RELATED REACTIONS

reaction to infusion	Preferred terminology	Preferred Term
	Infusion-related reactions	Infusion related reaction
	Infusion-associated hypersensitivity reactions	Infusion related hypersensitivity reaction
	Rapid onset allergic reaction	Anaphylactic reaction
	Rapid onset anaphylactic shock	Anaphylactic shock
	tachypnoea	Anaphylactoid reaction
	anaphylactic shock	Anaphylactoid shock
	hypersensitivity	Hypersensitivity
	Type I hypersensitivity	Type I hypersensitivity
	drug induced hypersensitivity	Drug hypersensitivity
	Rapid onset severe allergic reactions	Anaphylactic reaction
	flush	Flushing
	anaphylactic shock	Anaphylactic shock